Toward a metaplectic Jacquet–Langlands transfer, IV: Wild Local Correspondence and Global Trace Formula Comparison

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Abstract

This continuation of Parts I–III studies the two unresolved steps in a metaplectic Jacquet–Langlands transfer: integrality of the genuine character transform at places dividing the covering degree, and compatibility of the local transfer with a global trace-formula comparison. We formulate a precise integral-character-lattice criterion for the wild local problem. For the standard Kazhdan–Patterson diagonal-torus cocycle, we compute the commutator radical under the scalar-centrality hypothesis and obtain an explicit good-element criterion for regular split diagonal elements, at both tame and wild places. We explain why this does not yet settle the nonsplit-torus case, global geometric coefficients, or spectral separation. The latter questions are organized into verifiable proof obligations. No complete wild character correspondence or global Jacquet–Langlands theorem is claimed.

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1 Introduction: a continuation with explicit proof boundaries

The classical Jacquet–Langlands correspondence relates representations of $D^\times$ to essentially square-integrable representations of $\mathrm{GL}_r(F)$, where D/F is a central division algebra of degree r [1]. A metaplectic variant involves genuine representations of central extensions of these groups, and is not obtained merely by pulling classical representations back to the covering groups.

The preceding parts of this series develop (I) the specific covering cocycles, their centers, and good elements; (II) the local and global matching of orbital integrals, under stated hypotheses; and (III) a normalized character transform on the division-algebra side. In particular, Part III constructs an isometry

$$L_D : \mathcal{C}_\theta(D^\times) \longrightarrow \mathcal{C}_{\widetilde{\theta}}(\widetilde{D^\times}) \tag{1}$$

of invariant smooth functions with fixed central characters, proves a tame character result, and constructs an adjoint orbital-integral transfer, subject to the exact Part I cocycle and measure conventions. At wild places the corresponding assertion on individual irreducible characters remains open.

Notation. Locally F is a nonarchimedean field of characteristic zero containing μ_n, $n \geq 1$ is odd, $r \geq 2$ is prime, and $(r, n) = 1$. Put $H = D^\times$, $H^1 = \ker \mathrm{Nrd}$, $Z = F^\times$ and $\widetilde{H} = \widetilde{D^\times}$. A genuine character of a cover restricts to a prescribed faithful character of μ_n. We use the central-character and star-map conventions of Part III. When discussing $G = \mathrm{GL}_r(F)$, T denotes its diagonal torus and Z_G its scalar center. Globally K is a number field containing μ_n.

Mathematical status labels. **[E]** means imported from a cited source or an earlier part of the series; **[P]** is proved here under its explicit hypotheses; **[R]** is a proved *reduction*, not a solution of the original problem; **[O]** is open; and **[C]** is a conditional implication. The exact Part I cocycle and the associated measure normalizations have not been independently checked in this draft; Claude's proposed cross-part review is therefore a particularly valuable prerequisite before making claims stronger than those below.

Topic	Result in this draft	Status
Wild $D^\times$ character integrality	Character-lattice criterion and computable coefficients	[R]
Split diagonal-torus good elements	Exact radical under the specified torus cocycle	[P]
Good elements for nonsplit tori	Precise remaining local test	[O]
Global power classes modulo scalars	Almost-everywhere Hasse principle, recalled from III	[E]
Global trace-formula coefficients	Measure and orbit accounting checklist	[O]
Spectral separation and the local JL map	Conditional route, not a bijection	[C]

2 Wild-place character integrality as an integral-lattice problem

At residue characteristic $p \mid n$, Part III gives a unitary transformation of invariant class functions but does not establish that it maps every irreducible character to a genuine irreducible character. Here is a precise way to isolate that missing assertion.

2.1 The character matrix

Fix the central character θ of H and the corresponding genuine central character $\widetilde{\theta}$ of $\widetilde{H}$. Let

$$\mathcal{R}_\theta(H) = \bigoplus_{\rho \in \mathrm{Irr}_\theta(H)} \mathbb{Z}\Theta_\rho, \qquad \mathcal{R}_{\widetilde{\theta}}(\widetilde{H}) = \bigoplus_{\tau \in \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})} \mathbb{Z}\Theta_\tau$$

be the *integral virtual-character lattices*; the direct sums mean finite integral combinations of characters. For fixed unitary central character, Schur orthogonality on the compact-mod-center groups makes these irreducible characters orthonormal bases of the respective invariant L^2 spaces.

Write

$$a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau \rangle_{\widetilde{H}/\widetilde{Z}}, \qquad L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau. \tag{2}$$

The coefficients are *complex* a priori. With the regularity and smoothness assertions of Part III, each expansion is finite, but this finiteness should not be confused with integrality.

Proposition 2.1 ([**R**] The integral-character-lattice criterion)**.** *Assume the unitarity, smoothness, and finite character expansions of Part III. The following are equivalent:*

(i) L_D *carries each irreducible character of* H *to an irreducible genuine character of* $\widetilde{H}$ *and induces a bijection on their sets;*

(ii) $L_D\mathcal{R}_\theta(H) = \mathcal{R}_{\widetilde{\theta}}(\widetilde{H})$*;*

(iii) every coefficient $a_{\rho\tau}$ *and every coefficient of* $L_D^{-1}\Theta_\tau$ *is an integer.*

In (iii) it is enough to prove integrality of the coefficients $a_{\rho\tau}$ *if the inverse matrix is the adjoint matrix and its columns are the same finite families; then the inverse coefficients are complex conjugates of integers and hence are integers. The positivity at the identity specified by Part III determines the positive sign in the resulting signed permutations.*

Proof. (i) implies (ii) and (iii). Condition (ii) says that L_D and its inverse preserve the indicated integral bases, and therefore implies (iii). Under (iii), the ℓ^2-norm of every row of the unitary matrix $A = (a_{\rho\tau})$ is one. A finitely supported integral row of norm one has exactly one nonzero entry, equal to $+1$ or -1. The same argument for the inverse implies that these assignments produce a bijection of the bases. Finally, the identity-value formula of Part III gives a strictly positive value for $L_D\Theta_\rho$ at the identity, so the sign is positive. □

Remark 2.2 (Why the inverse condition matters)**.** For an arbitrary map between free abelian groups, mapping a basis to integral combinations need not imply surjectivity. Here unitarity, together with the basis expansion, is crucial: an integral unit vector is already a signed basis vector. The additional inverse statement ensures that no target irreducible is omitted.

2.2 Numerical necessary conditions and a finite computation

Part III identifies the positive normalization

$$c_D = |n|_F^{-(r^2-1)/2}, \qquad (L_D\Theta_\rho)(1) = c_D \dim \rho. \tag{3}$$

Accordingly, if $L_D\Theta_\rho = \Theta_\tau$, then

$$\dim \tau = c_D \dim \rho. \tag{4}$$

This condition is necessary, not sufficient. For example, at $F = \mathbb{Q}_3(\zeta_3)$, $r = 2$, $n = 3$, one has $|3|_F = 3^{-2}$, so $c_D = 27$. The dimension of a purported image must be divisible by 27. The covered-torus argument in Part III supplies only a weaker factor of 3 for every genuine irreducible; it does not supply an example that violates the stronger factor.

Proposition 2.3 ([**R**] Finite-level test for a fixed representation)**.** *Fix $\rho \in \mathrm{Irr}_\theta(H)$ and assume the smoothness assertion for $L_D\Theta_\rho$ from Part III. Only finitely many $a_{\rho\tau}$ can be nonzero, and integrality of the image character is decided by the finite collection of equalities*

$$a_{\rho\tau} \in \mathbb{Z} \qquad (a_{\rho\tau} \neq 0).$$

Each coefficient can be computed by finite character sums after choosing sufficiently small compact open normal subgroups compatible with the given central characters.

Proof. A smooth invariant function with fixed central character on a compact-mod-center group is determined at a sufficiently small normal level of the resulting compact projective quotient. Its character expansion lies in the finite character space of that finite twisted quotient. Integration against the normalized Haar measure becomes a finite sum on this quotient. The coefficient formula is exactly (2). □

Remark 2.4 (A falsifiable research direction)**.** A single ρ for which one coefficient in (2) is nonintegral disproves irreducibility preservation by the *specified* L_D. It need not disprove the existence of another wild local correspondence. Conversely, checking finitely many representations at finitely many levels does not prove the statement at every depth.

2.3 Why the tame Adams proof cannot be repeated

For a finite group Q, the Adams operation satisfies $\psi^n\chi(g) = \chi(g^n)$. When n is prime to $|Q|$, cyclotomic Galois conjugation makes this a permutation of irreducible characters; this is the ingredient used in the tame case of Part III. It fails when $p \mid n$ and Q has p-torsion.

Remark 2.5 (An elementary obstruction)**.** For $Q = C_p$ and $n = p$, every $g \in Q$ satisfies $g^p = 1$, hence $\psi^p\chi(g) = \chi(1)$. In particular all linear characters have the same Adams image, the trivial character. Thus the tame Adams argument is not available at a wild place without new structure.

3 A split-place theorem: the commutator radical

The preceding discussion concerns the division side. A second bottleneck is to characterize *good* regular elements of the metaplectic cover at places where the division algebra is split. The first case, regular elements in the split diagonal torus, admits an exact calculation.

3.1 The standard torus cocycle and its matrix

Let $G = \mathrm{GL}_r(F)$, with diagonal torus $T \simeq (F^\times)^r$ and scalar center $Z_G \simeq F^\times$. Assume the restriction of the chosen cocycle to T is represented by the standard Kazhdan–Patterson convention

$$\sigma_c(t,u) = (\det t, \det u)_n^c \prod_{1\le i<j\le r} (t_i,u_j)_n, \quad t = \mathrm{diag}(t_i), \quad u = \mathrm{diag}(u_i). \tag{5}$$

Here $(\cdot,\cdot)_n$ is the nth Hilbert symbol on F. This normalization is compatible with the scalar-centrality congruence used in Part I,

$$r - 1 + 2rc \equiv 0 \pmod{n}. \tag{6}$$

Equivalent cocycles give the same commutator pairing, but before using this calculation globally the convention (5) needs to be checked against the exact local section used in Part I. The Kazhdan–Patterson and later block-compatible formulas are discussed in [2, 3, 4]. The simple-type results in [4] assume $(p, n) = 1$ and therefore do not resolve the present wild problem.

Since n is odd, the Hilbert symbol is alternating: $(x, y)_n = (y, x)_n^{-1}$. Directly taking the commutator of the two cocycle lifts gives

$$B_c(t, u) := [\widetilde{t}, \widetilde{u}] = \prod_{i,j=1}^{r} (t_i, u_j)_n^{A_{ij}}, \quad A = -I_r + (1 + 2c)J_r, \tag{7}$$

where J_r is the all-ones matrix. In particular the diagonal entries of A are $2c$ and the off-diagonal entries are $1 + 2c$. Its row and column sums equal $r - 1 + 2rc$, which vanish modulo n under (6); accordingly the scalar lifts commute with $\widetilde{T}$.

Lemma 3.1 ([**P**] Radical of the commutator matrix)**.** *Assume n is odd and $(r, n) = 1$, and let $X = F^\times/F^{\times n}$. The bilinear pairing*

$$X^r \times X^r \longrightarrow \mu_n, \qquad (x, y) \longmapsto \prod_{i,j} (x_i, y_j)_n^{A_{ij}}$$

with A as in (7) and subject to (6) has radical exactly the diagonal subgroup $\Delta(X) = \{(a, \dots, a) : a \in X\}$.

Proof. The Hilbert pairing $X \times X \to \mu_n$ is perfect by local Kummer theory and local duality. Hence the radical of the displayed pairing is precisely $\ker(A : X^r \to X^r)$. The group X is killed by n. Because $(r, n) = 1$, multiplication by r is an automorphism of X, and every $x \in X^r$ decomposes uniquely as

$$x = (a, \dots, a) + w, \qquad \sum_i w_i = 0.$$

On the diagonal summand, A acts by $r - 1 + 2rc \equiv 0 \pmod{n}$. On the sum-zero summand it acts as $-I_r$, since $J_r w = 0$. Thus it is invertible on that summand and its kernel is precisely $\Delta(X)$. □

Theorem 3.2 ([**P**] Good regular elements of the split diagonal torus)**.** *Let $t = \mathrm{diag}(t_1, \dots, t_r)$ be regular in $G = \mathrm{GL}_r(F)$. Assume (5) and (6), with n odd and $(r, n) = 1$. A lift $\widetilde{t}$ is good, in the sense that it commutes with the full preimage of $G_t = T$, if and only if*

$$t \in Z_G T^n. \tag{8}$$

Equivalently,

$$t_i/t_j \in F^{\times n} \quad \text{for all } i, j. \tag{9}$$

This criterion applies at both tame and wild places.

Proof. For regular t, its centralizer in G is T. By the definition of good elements, $\widetilde{t}$ is good if and only if its commutator with every lift of $u \in T$ is trivial. In view of (7), this is exactly membership in the radical of the torus pairing. Lemma 3.1 identifies the radical on X^r with the diagonal. The inverse image of this subgroup in T is $Z_G T^n$. Condition (9) is equivalent to that assertion, since t_i/t_1 being an nth power allows one to factor t into a scalar and a coordinatewise nth power. □

Remark 3.3 (The star-map description)**.** Under the scalar-centrality hypothesis, a regular good diagonal lift is a central lift times the nth power of a lift of another diagonal element. Thus the result matches the star-map description used on the division side in Part III; it does not require the nth-power map to be an automorphism of T.

Remark 3.4 (Concrete example: $r = 2$, $n = 3$)**.** Take $F = \mathbb{Q}_3(\zeta_3)$ and $c \equiv 2 \pmod 3$. Then $1 + 4c \equiv 0 \pmod 3$ and

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3, \qquad B_c(t,u) = (t_1/t_2, u_1/u_2)_3.$$

Perfection of the cubic Hilbert pairing shows that a regular $\mathrm{diag}(t_1, t_2)$ is good exactly when $t_1/t_2 \in F^{\times 3}$; the residue characteristic dividing n does not alter this equivalence.

3.2 Exactly what the split calculation does not establish

A *split place for* D means $D \otimes_K K_v \simeq M_r(K_v)$; it does not mean that every elliptic or regular maximal torus in $\mathrm{GL}_r(K_v)$ is split. The centralizer of a regular element can be $\mathrm{Res}_{E_v/K_v} \mathbb{G}_m$ for a *field* or a product of fields E_v, not necessarily the diagonal torus. Theorem 3.2 therefore does not by itself identify all good regular classes in the local split group.

Definition 3.5 (**[O]** Nonsplit-torus problem)**.** For each degree-r *étale* F-algebra E embedded as a regular maximal torus $T_E \subset \mathrm{GL}_r$, compute the radical of the commutator pairing on $p^{-1}(T_E(F))$, with the exact local cocycle of Part I. Determine whether it is the preimage of $F^\times T_E(F)^n$, or whether additional local norm or cohomological classes occur. The answer must be uniform in E and include the places where $p \mid n$.

A possible route is to express the commutator via restriction and corestriction of Hilbert symbols on the factors of E. At a split place this requires a computation on each rational torus, not merely a calculation after extending scalars to an algebraic closure. A formula valid after scalar extension need not automatically imply an equality of radicals on F-rational points.

4 Global power classes, rational orbits, and their limits

The split diagonal calculation suggests a connection between good elements and nth powers modulo scalars. Part III established a useful global arithmetic result in precisely this direction.

Proposition 4.1 (**[E]** Almost-everywhere Hasse principle, Part III)**.** *Let E/K be a field extension of degree r, let $\mu_n \subset K$, and assume $(r, n) = 1$. For any finite set Σ of places of K, the localization map*

$$\frac{E^\times}{K^\times E^{\times n}} \longrightarrow \prod_{v \notin \Sigma} \frac{(E \otimes_K K_v)^\times}{K_v^\times (E \otimes_K K_v)^{\times n}} \tag{10}$$

is injective.

Proof recalled for completeness. Suppose the class of $t \in E^\times$ is trivial in every local factor outside Σ. For such v write $t = a_v y_v^n$ with $a_v \in K_v^\times$ and $y_v \in (E \otimes_K K_v)^\times$. Choose integers u, k such that $ru + nk = 1$. Put $a = \mathrm{N}_{E/K}(t)^u \in K^\times$. Taking local norms and rearranging gives

$$\frac{t}{a} = \left(a_v^k \frac{y_v}{\mathrm{N}(y_v)^u} \right)^n.$$

Thus t/a is an nth power in every completion of E above $v \notin \Sigma$. If t/a were not an nth power in E, the Kummer extension $E((t/a)^{1/n})/E$ would be nontrivial because $\mu_n \subset E$. Chebotarev would then give infinitely many unramified places of E at which t/a is not a local nth power, a contradiction. Hence $t \in K^\times E^{\times n}$. □

Remark 4.2 (No automatic global cover comparison)**.** The proposition tests local powers modulo scalars at *almost all places*. It cannot be invoked from the good-element condition unless that condition has first been identified with the local power condition for the *actual* regular centralizer at those places. Theorem 3.2 proves one case, not the needed uniform statement. Global reciprocity for the covering cocycle and compatibility of adelic star lifts are additional questions.

4.1 What Kottwitz's framework already resolves

Let D/K be a central simple algebra of degree r, and let a regular elliptic element correspond to a degree-r field E/K. Its centralizer as an algebraic group is the induced torus $T = \mathrm{Res}_{E/K}\,\mathbb{G}_m$. Hilbert's Theorem 90 and Shapiro's lemma give $H^1(K,T) = 1$, so a represented stable regular class does not split into multiple rational conjugacy classes. Existence of an embedding $E \hookrightarrow D$ is equivalent to $D \otimes_K E$ being split; the Brauer–Hasse–Noether theorem detects this condition locally.

These are special simplifications, in the inner forms of GL_r, of the rational-versus-stable orbit and torus-transfer formalism of Kottwitz [5, §§7–9]. His formula for relative Tamagawa numbers [5, §5] supplies the correct context for global volume factors. None of these linear-group statements by itself controls genuine conjugacy, the star map, or the adelic covering cocycle.

5 Geometric normalization: the next exact comparison

The trace formulas cannot be compared by multiplying the local probability-measure constants from Part III without examining their relation to the globally fixed Haar measures.

5.1 Three distinct quantities

At a division place v, Part III introduces

$$b_{D,v} = |n|_v^{-(r-1)/2} \quad \text{and} \quad c_{D,v} = |n|_v^{-(r^2-1)/2}.$$

The former is the normalized elliptic class-function/orbital-integral factor for the *local probability quotient measure*; the latter is its value-at-identity scaling. Neither is automatically the global geometric factor. A third contribution is the conversion between those local probability measures and the Haar measures used in the adelic trace formula. These three quantities must remain separate in every local comparison.

The product formula $\prod_v |n|_v = 1$ is true for $n \in K^\times$, but does not by itself cancel the $b_{D,v}$: these constants occur at the division places, whereas the complement includes split and archimedean places with their own transfer factors.

5.2 An explicit normalization ledger

For each rational regular elliptic class γ, the desired geometric identity should be checked term by term with the following data, using one globally fixed normalization:

Input	Required verification
Local orbit	Is the lift good? Does the rational stable class have a matching lift?
Local orbital integral	Exact transfer factor, sign, power-map multiplicity, and Jacobian
Local quotient measure	Conversion of T_v/Z_v probability measure to the adelic Haar measure
Global torus volume	$T(K)\backslash T(\mathbb{A}_K)$ factor with the center and connected components treated consistently
Global cocycle	Product/reciprocity law for the chosen lifts; dependence on c_v
Singular and identity terms	Correct treatment beyond regular elliptic classes, or a test-function hypothesis eliminating them
Spectral side	Absolute convergence or finite packet identity and a justified separation argument

Definition 5.1 ([**O**] Geometric comparison problem)**.** Fix global measures and matched factorizable genuine test functions. Prove a trace-formula identity by identifying the regular elliptic geometric terms, including their global centralizer volumes and all local factors, and then handling or eliminating singular terms. The comparison is *not* asserted by the existence of local transfer operators alone.

Remark 5.2 (Available global trace formula)**.** Wen-Wei Li constructed an invariant trace formula for adelic covers of connected reductive groups under specified hypotheses, including relevant covers of GL_r [6]. This is an important global tool, but it must be specialized to the present cocycles and matched functions. A theorem about the existence of a trace formula is not the same as the required equality of the two geometric sides.

6 Spectral separation and a proposed order of attack

Even a geometric identity does not automatically produce a bijection between local irreducible representations. A global spectral comparison will yield combinations of local characters, possibly with multiplicities. To recover an *individual* character, one needs a separation principle, a supply of suitable globalizations, and control of every other local factor.

Proposition 6.1 ([**R**] What a spectral identity must detect)**.** *Let $\rho \in \mathrm{Irr}_\theta(D_v^\times)$ be fixed at a wild place. Suppose a proved global identity, using test functions varied at v, implies the exact equality of local invariant distributions*

$$L_D\Theta_\rho = \sum_\tau m_\tau \Theta_\tau, \qquad m_\tau \in \mathbb{Z},$$

with only finitely many terms and the normalization of Part III. Then $L_D\Theta_\rho$ equals a single irreducible genuine character. If the same conclusion holds for all ρ and the inverse coefficients are integral, the local wild correspondence is a bijection.

Proof. This is Proposition 2.1, applied to one row of the unitary character matrix. The hypothesis that the spectral identity supplies *integer* coefficients is the essential additional input: neither a geometric identity nor L^2 unitarity establishes it without an argument. □

Preferred order of work. The next rigorous advance should be the nonsplit regular-torus commutator calculation, followed by the unramified transfer and the adelic normalization ledger. In parallel, the finite-level computation of (2) for the smallest wild example $(r, n, F) = (2, 3, \mathbb{Q}_3(\zeta_3))$ is a useful independent test: one nonintegral coefficient would change the strategy immediately. Only after these checks should the spectral comparison be promoted to a claimed theorem. Removal of the auxiliary place and global non-cancellation hypotheses belongs after a correct base trace identity is available.

7 Glossary of specialized terms

Adams operation

The operation ψ^n on a representation ring, characterized on characters of a finite group by $\psi^n\chi(g) = \chi(g^n)$. It is related to exterior powers in equivariant K-theory, and may produce virtual rather than irreducible characters.

Character lattice

The free abelian group generated by irreducible characters with a fixed central character; its elements are integral virtual characters.

Good element

A lift whose commutator with the full preimage of the ordinary centralizer is trivial. A genuine invariant distribution is supported on the appropriate good locus.

Hilbert symbol

The local norm-residue pairing $(\cdot,\cdot)_n$ on $F^\times/F^{\times n}$, valued in μ_n. It is perfect when $\mu_n \subset F$.

Commutator radical

The subgroup of a torus on which the lifted commutator pairing is trivial against every element of that torus.

Split place / split torus

A split place of D is one where $D_v \simeq M_r(K_v)$; a split torus is a torus isomorphic over K_v to a product of copies of $\mathbb{G}_m$. These are different notions.

Central character

A prescribed scalar by which the center acts in an irreducible representation. Compatibility of these characters is required when comparing representations of different covers.

Integral virtual character

A finite integer combination of irreducible characters, not necessarily the character of an actual representation.

Rational and stable conjugacy

Conjugacy under $G(K)$ and under $G(\overline{K})$, respectively. Their difference is measured by Galois cohomology of centralizers.

Induced torus

For E/K finite étale, $\mathrm{Res}_{E/K}\,\mathbb{G}_m$; its K-points are $E^\times$ and its first Galois cohomology vanishes by Shapiro and Hilbert 90.

Kottwitz obstruction

A cohomological obstruction to realizing locally transferred maximal tori in a global inner form; in Kottwitz's setting it is expressed using $H^2(K, T_{\mathrm{sc}})$.

Tamagawa measure

The canonical global measure built from compatible local algebraic measures and convergence factors. It need not agree with probability Haar measure on a compact local quotient.

Kummer class
: A class in $E^\times/E^{\times n}$ or its torus analogue; Kummer theory converts it into a finite Galois-cohomology class.

Spectral separation
: The deduction of individual local character identities from a global equality of sums of automorphic representation traces.

Local-to-global obstruction
: A class that disappears in every localization but remains nontrivial globally. The power-class kernel recalled here vanishes under the stated hypotheses; covering-group compatibility may have additional obstructions.

Acknowledgments and verification scope

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